

Deep learning-guided dual-fitness evolution of T7 RNA polymerase for enhanced stability and activity

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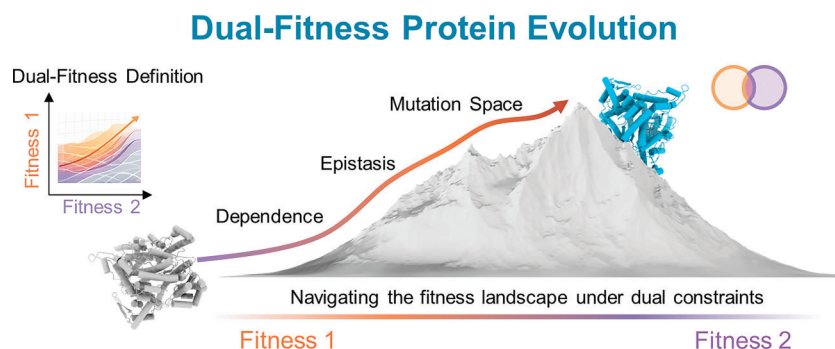
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Abstract

In protein engineering, simultaneously improving multiple fitness attributes is a critical yet challenging goal, largely due to the vastness of sequence space, the multifaceted interplay among different traits, and the complexity of non-linear mutational effects (epistasis). To address this, we developed a data-driven evolutionary strategy that couples *in silico* deep learning with a wet-lab multi-objective selection workflow. By employing independent model fine-tuning for distinct traits, our approach facilitates navigating the fitness landscape to identify beneficial mutation combinations. We applied this strategy to T7 RNA polymerase (T7 RNAP), performing dual-fitness evolution to simultaneously enhance thermostability and activity at elevated temperatures. After five rounds of iterative evolution, we obtained T7 RNAP mutants exhibiting a melting temperature (T_m) increase of $>10^\circ\text{C}$, a 60-fold enhancement in high-temperature activity, and a 70% reduction in by-product content. Validation in cell transfection demonstrated their potential for producing high-quality mRNA for industrial applications.

Graphical abstract



Introduction

Protein engineering is driven by the need to improve the functionality, stability, and efficiency of proteins to meet diverse application demands. While substantial progress has been made in single-fitness protein engineering [1–4], real-world applications often necessitate the concurrent optimization of multiple fitness objectives.

Multi-fitness co-evolution faces two primary challenges. The first is the complex dependencies between different traits, such as trade-off, functional thresholds, or weak correlations. A trade-off occurs when the improvement of one fitness trait comes at the expense of another [5–7]. A well-documented example is the tension between protein stability and catalytic

activity [8]. Specifically, catalytic activity usually requires conformational flexibility to enable substrate binding and transition state stabilization, yet such structural flexibility often destabilizes the protein fold [9]. Notably, this trade-off typically pertains to activity at ambient temperatures. The relationship between thermostability and higher temperature activity is distinct since increased thermostability is necessary but not sufficient to improve the activity at high temperature. Furthermore, studies have indicated that other traits may exhibit weak correlations, such as the independent relationship observed between thermostability and acid resistance [10]. The second challenge arises from epistatic effects—non-linear interactions between mutations where the combined effect de-

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viates from the simple sum of individual contributions [11, 12]. Such an interaction complicates the predictions of multi-site mutation outcomes based solely on single-site mutants [13, 14].

Conventional methods, such as stepwise greedy algorithms, typically build upon top-performing mutants from prior rounds [15–17]. While effective in local searches, these strategies are inherently constrained by their limited exploration of the vast sequence space and may struggle to identify beneficial combinations that require crossing fitness valleys [18, 19]. These limitations underscore the need for efficient, data-driven strategies capable of navigating complex fitness landscapes.

With the rapid advancement of deep learning algorithms, there is growing potential to quickly extract features from vast and complex datasets [20–24]. Leveraging the extensive data accumulated over the past decades on protein mutation and next-generation sequencing, deep learning holds promise for accelerating the progress in protein engineering [25]. Pre-trained protein language models (PLMs) can predict the probability distribution of amino acid mutations, thereby identifying the most naturally occurring variants [22, 26]. Although these predictions are initially undirected, incorporating functional data can steer them toward targeted properties. Fine-tuning PLMs with experimental data further narrows the scope, enabling more focused optimization of the fitness of a specific protein [2]. Yet, most existing studies predominantly focus on single-fitness optimization or the linear combination of single-site mutations [27, 28], leaving substantial opportunities unexplored in multi-fitness protein evolution, particularly regarding epistatic fitness landscapes where non-additive combinations of mutations can outperform individually beneficial substitutions.

T7 RNA polymerase (T7 RNAP) is a cornerstone of molecular biology and biotechnology, serving as a primary candidate for multi-objective engineering given its pivotal role in *in vitro* transcription [29–31], mRNA vaccine synthesis [32–35], and isothermal amplification [36–38]. Despite its versatility, a persistent bottleneck is its intrinsic thermostability. Wild-type (WT) T7 RNAP begins to denature at $\sim 40^\circ\text{C}$, resulting in a precipitous loss of catalytic activity [39, 40]. This thermal sensitivity severely constrains its industrial application. For instance, operating temperatures above 48°C are required to suppress the formation of immunogenic double-stranded RNA (dsRNA), which complicates mRNA vaccine purification and triggers adverse immune responses [41–45]. Furthermore, isothermal techniques such as nucleic acid sequence-based amplification (NASBA) [36] and transcription-mediated amplification (TMA) [37] demand elevated temperatures ($\geq 41^\circ\text{C}$) to resolve inhibitory RNA secondary structures [46]. Crucially, simply increasing thermostability does not always translate to robust high-temperature activity. Consequently, the simultaneous enhancement of both thermal stability and high-temperature performance is vital to unlocking the full industrial potential of T7 RNAP.

Here, we develop a dual-fitness deep learning-guided evolution strategy based on a PLM [2], emphasizing the systematic evaluation of multi-site combinational mutations. This approach synergistically integrates evolutionary insights from pre-trained PLMs with experimental feedback to prioritize high-potential variants, thereby addressing the generalization limitations of purely *in silico* predictions. Specifically, we fine-

tuned separate PLMs for two distinct fitness attributes and designed a combinatorial selection pipeline that ranks variants based on one attribute while imposing a cut-off threshold for the other. We applied this strategy to guide the simultaneous evolution of thermostability and high-temperature activity for T7 RNAP. Through successive rounds of *in silico* prediction and *in vitro* validation, we obtained several mutants with melting temperature (T_m) exceeding that of the WT by 10°C and high-temperature activities surpassing those of the WT by >60 -fold. Furthermore, validation in cell transfection assays confirmed that the stable, high-activity variant notably reduced dsRNA by-product content by $\sim 70\%$, demonstrating their practical utility for high-quality mRNA production.

Materials and methods

Materials

All plasmids and iSpinach DNA templates were synthesized by Beijing Tsingke Biotech, Sangon Biotech, and Yaohaibio. The Quant-iT RiboGreen RNA kit was purchased from Invitrogen. DNase I, the Monarch PCR & DNA Cleanup Kit, and the Monarch RNA Cleanup Kit were obtained from New England Biolabs (NEB). Restriction enzymes, NTP solutions, RNase inhibitors, and the plasmid extraction kit were sourced from Sangon Biotech. The Cap1 capping kit and dsRNA ELISA (enzyme-linked immunosorbent assay) kit were acquired from Hzymes Biotech. DFHBI (3,5-difluoro-4-hydroxybenzylidene imidazolinone) was purchased from Topscience.

In silico mutation predictions

We chose the PLM PRIME for zero-shot and supervised mutant scoring [2]. PRIME is a transformer-based protein language model pre-trained on masked language modeling and optimal growth temperature (OGT) prediction tasks. It consisted of three main modules: an encoder, a masked language model (MLM) module, and an OGT predictor. The model integrated protein sequences with the optimal growth temperature of their host organisms during training. This allowed it to predict the stability and activity of protein single-site mutants in a zero-shot manner, without any prior experimental mutagenesis data. The model could also be fine-tuned with experimental data to predict multi-site mutants. More details can be found in a previous study [2] and the Supplementary data.

Zero-shot mutant scoring

Like other protein language models [26], PRIME could score the mutant using the log odds of the mutant and the WT: $\log p(x_i = m_{AA}|w) - \log p(x_i = w_{AA}|w)$, where w represents the one-hot encodings of the WT, x_i denotes the amino acid of position i , m_{AA} is the mutation type, and w_{AA} is the amino acid of the WT. The scoring did not need any prior labeled data of mutants.

Supervised mutant fitness prediction

We utilized the PRIME encoder to map mutant sequences into the latent space and employed an additional regression module to learn the mapping relationship between the latent representations and their specific properties. This regression module was implemented as a two-layer multi-layer perceptron (MLP; a neural network with two hidden layers). The complete architecture was trained to minimize the mean square

error between predicted scores and experimentally measured fitness for T7 RNAP.

Dual-fitness prediction and selection

We employed a full fine-tuning method to train two separate models—one for stability and another for activity—using the pre-trained encoder and regression head. These two models were then employed to score all combinatorial multi-site mutants. The multi-site variants were generated as combinations of the Top-N single-site mutants obtained in Step 1, with scores standardized using the z-score method. We ranked these multi-site mutants based on predicted activity and used predicted stability as a cut-off to filter the sequences, resulting in Mutation Set 1. Similarly, we ranked the mutants based on predicted stability and used the predicted activity as a cut-off to filter the sequences, resulting in Mutation Set 2. The union of Set 1 and Set 2 was selected as the candidate for experimental testing.

Iterative multi-site mutant selection

The selection process operated iteratively and concluded once a multi-site mutant meeting the desired criteria was identified. If no such mutant was identified, experimentally labeled mutants, including both negative and positive data, were added to the training set to fine-tune the models. The updated models were then used to select new mutant candidates. This iterative cycle persisted until an ideal mutant was obtained.

Protein expression and purification

The T7 RNAP gene was cloned into the pQE-80L expression vector and transformed into *Escherichia coli* BL21(DE3) cells. The plasmid map of pQE-80L is shown in [Supplementary Fig. S1](#). The cells were cultured in LB medium until reaching an OD₆₀₀ of ~0.6–0.8, followed by induction with 1 mM isopropyl-β-D-thiogalactopyranoside (IPTG) for a 6 h growth period at 37°C. After collection, the bacteria were resuspended in a binding buffer (50 mM Tris-HCl pH 8.0, 300 mM NaCl, 3 mM imidazole, 0.1 mM EDTA) and lysed via sonication. The resulting lysate underwent centrifugation at 4°C and 12 000 rpm for 30 min. The lysate was then applied to an Ni-NTA gravity column and washed with a washing buffer (50 mM Tris-HCl pH 8.0, 300 mM NaCl, 10 mM imidazole, 0.1 mM EDTA, 10% glycerol). Elution was performed using an elution buffer (50 mM Tris-HCl pH 8.0, 300 mM NaCl, 250 mM imidazole, 0.1 mM EDTA, 10% glycerol). Concentration was achieved using a final ultrafiltration buffer (50 mM Tris-HCl pH 8.0, 100 mM NaCl, 0.1 mM EDTA), and the T7 RNAP was diluted with a storage buffer [50 mM Tris-HCl pH 8.0, 100 mM NaCl, 0.1 mM EDTA, 1 mM dithiothreitol (DTT), 75% glycerol] [47, 48].

Differential scanning fluorimetry

The protein staining agent SYPRO Orange (5000× stock) was diluted to 5× in the reaction system and mixed with the protein sample (0.2 mg ml⁻¹) in an eight-row PCR tube. Each sample was prepared in a final volume of 20 μl and tested in triplicate. Denaturation curves were generated using a PCR instrument (Analytik Jena qTower3) equipped with appropriate optical filters (FAM 470 nm and ROX 625 nm for excitation and emission, respectively). The temperature was incrementally increased by 0.5°C steps from 25 to 65°C, with a 5 s hold for equilibration at each temperature step. The thermal

unfolding curves were analyzed by fitting the Boltzmann equation to approximate the T_m [49].

Circular dichroism

Far-UV circular dichroism (CD) experiments were carried out with a JASCO J-1500 spectrometer. The thermal inactivation experiment was detected at a wavelength of 220 nm, and the temperature was increased from 25 to 65°C in steps of 1°C min⁻¹ with 30 s of equilibration time. The unfolding fraction, f_u , was calculated using $f_u = (CD_T - CD_{min}) / (CD_{max} - CD_{min}) \times 100\%$, where CD_T is the observed CD value at a certain temperature, and CD_{max} and CD_{min} are the maximum and minimum CD values of the thermal inactivation process of proteins [50].

Differential scanning calorimetry

Differential scanning calorimetry (DSC) experiments were performed with a nanoDSC instrument from TA company. The WT and mutants were all dissolved in phosphate-buffered saline (PBS) solution (pH 7.5) at a protein concentration of 1 mg ml⁻¹. The reference was pure PBS solution under the same conditions. Both sample and reference were degassed at 4°C for 15 min to ensure that there were no bubbles in the solutions. The pressure of the instrument chamber was set to 3 bar, and the scanning was performed during the heating process from 25 to 65°C at a rate of 1°C min⁻¹ after 600 s equilibration. The reference–reference signal was subtracted as the baseline from the sample–reference scanned. The curve was modeled using a two-state scaled model [51].

In vitro transcription assay

The *in vitro* transcription (IVT) reaction buffer was prepared, which contained 200 mM HEPES (pH 7.5), 30 mM MgCl₂, 20 mM DTT, 0.4 U μl⁻¹ RNase inhibitor, 5 mM NTP mix, and 100 nM iSpinach DNA template. The buffer was incubated at 37 and 52°C for 10 min and then 0.04 mg ml⁻¹ T7 RNAP was added to initiate the reaction. After the mixture was incubated for 1 h, 0.5 M EDTA was added to stop the reaction. When screening the ideal mutants, the reactions were carried out at 52 and 55°C for 3 h, respectively. Finally, 100 μM DFHBI was introduced, and fluorescence was measured with excitation at a wavelength of 470 nm and emission at 512 nm [47]. Each test included a blank control, with the WT serving as the reference. Unless otherwise specified, all wet-lab experiments described hereafter were performed in triplicate.

To investigate the impact of temperature on the activity of T7 RNAP with a short template, the IVT reaction buffer was incubated at temperatures of 37, 40, 42.5, 45, 47.5, 50, 52.5, 55, and 57.5°C for 10 min. T7 RNAP was then added and the reaction was allowed to proceed for 30 min and terminated by 100 mM EDTA. The intensity of the aptamer fluorescence was measured to analyze the effects of temperature.

The testing of T_{50}^{15} involved incubating both the WT and the mutant variants at temperatures of 40, 42.5, 45, 47.5, 50, 52.5, 55, 57.5, 60, and 62.5°C for 15 min. Subsequently, their residual transcriptional activity was measured at 37°C for 30 min.

mRNA synthesis and characterization

Templates of ~1000, ~2000, and ~4000 bp encoding enhanced green fluorescent protein (EGFP), luciferase, and Cas9

sequences were cloned into pTnTR and pET-28a(+) vectors and transformed into JM108 and DH5 α strains with ampicillin and kanamycin resistance, respectively. The plasmid maps of pTnTR and pET-28a(+) are shown in [Supplementary Figs S2 and S3](#). The transformed cells were cultured overnight at 37°C in LB medium containing the appropriate antibiotics. Plasmids were extracted using a plasmid extraction kit and linearized with BamHI for pTnTR vectors and XhoI for pET-28a vectors. The linearized plasmids were then purified using the Monarch PCR & DNA Cleanup Kit (NEB) to serve as templates for IVT. The IVT reaction mixture, with a total volume of 20 μ l, included 40 mM Tris-HCl (pH 7.5), 20 mM MgCl₂, 50 mM NaCl, 2 mM spermidine, 1 mM DTT, 5 mM NTP mix, 1 U μ l⁻¹ RNase inhibitor, 0.01% Triton X-100, 0.1 mg ml⁻¹ T7 RNAP (either WT or a heat-resistant mutant), and 300 ng of DNA template. WT reactions were conducted at 37°C, and mutant reactions at 50°C. Additionally, to study the temperature's effect on RNA yield, the mixture was incubated at 37, 42.5, 45, 47.5, 50, 52.5, and 55°C for 30 min. After transcription, 2 μ l of DNase I (5 U μ l⁻¹) was added. The mixture was vortexed for 3 s and then incubated at 37°C for 30 min to digest the DNA template. The IVT reaction mixture was then purified using the Monarch RNA cleanup kits (NEB). For mRNA intended for transfection, a capping reaction was performed using the Cap1 capping kit (Hzymes Biotech), followed by a secondary purification with the Monarch RNA cleanup kits (NEB) [32]. The recovery rates of the Monarch RNA cleanup kit for mRNAs of varying lengths are shown in [Supplementary Table S1](#).

Production and characterization of mRNA-containing lipid nanoparticles

mRNA-lipid nanoparticle (LNP) complexes were produced by microfluidics (PeciGenome). Lipids were dissolved in ethanol at molar ratios of 50:10:38.5:1.5 (25 mM SM102, MCE; 5 mM 1,2-distearoyl-*sn*-glycero-3-phosphocholine 18:0 DSPC, Avanti; 19.25 mM cholesterol, Sigma; 0.75 mM DMG-PEG-2000, Avanti). The complexes self-assembled by mixing the lipid solution with an acidic aqueous mRNA solution (100 mM acetate, pH 4) at a 1:3 volume ratio in a Tesla micromixer. The LNPs were then recovered, resuspended in PBS, and processed through a diafiltration column (100 kDa cut-off Amicon Ultra-4 Centrifugal Filter Unit).

The hydrodynamic diameter of mRNA-LNP complexes was characterized by dynamic light scattering with the Omni Multi-angle Particle Size and High Sensitivity Zeta Potential Analyzer (Brookhaven). Quantification of mRNA encapsulation into LNPs was performed with the Quant-iT RiboGreen RNA kit (Invitrogen) according to the manufacturer's instructions [31, 52].

dsRNA enzyme-linked immunosorbent assay

dsRNA content was determined using the dsRNA ELISA kit from Hzymes Biotech. mRNA samples were diluted to 0.5 ng μ l⁻¹, and dsRNA standards were prepared in a gradient ranging from 1 to 0 pg μ l⁻¹ (1, 0.5, 0.25, 0.125, 0.0625, 0.0312, 0.0156, and 0 pg μ l⁻¹). Microplate wells coated with anti-dsRNA antibodies captured dsRNA from the samples. Biotinylated detection antibodies, streptavidin-conjugated horseradish peroxidase (HRP), and 3,3',5,5'-tetramethylbenzidine (TMB) were added sequentially. The reaction mixture turns blue under the catalysis of HRP and

shifts to yellow upon acid addition. The OD measured at 450 nm with a 650 nm reference is directly proportional to the dsRNA content in the sample. The dsRNA concentration in the samples is calculated based on the dsRNA standard curve [41].

Cell culture

HEK293T cells were cultured in high-glucose Dulbecco's modified Eagle's medium (DMEM) with 1% penicillin-streptomycin, and 10% serum at 37°C and 5% CO₂. The cells were seeded in a 24-well plate at 150 000 cells per well and incubated for 24 h. Cell counts were performed using a Thermo Fisher cell counter. Luciferase-mRNA-loaded LNPs were then transfected at 5 μ g per well in triplicate and cultured for another 24 h. Culture medium was removed, and cells were lysed sequentially in each well for 5 min before transferring lysates to 1.5 ml centrifuge tubes. After centrifugation at 10 000 rpm for 10 min, the supernatant was collected for analysis. Luciferase expression was measured using luciferase assay reagent (Beyotime) with a SpectraMax iD5 (Molecular Devices) in luminescence mode [31, 53].

Molecular dynamics simulation

To investigate the origins of the changes in activity and thermostability, two sets of molecular dynamics (MD) simulations were performed. To explore activity changes, we carried out MD simulations of the protein-nucleic acid complexes in both the initiation and elongation conformations. Homology modeling of the WT and mutant proteins was performed based on the structures of the initiation complex (PDB ID: 1QLN) [54] and the elongation complex (PDB ID: 1MSW) [55]. These simulations employed the AMBER99SB force field with PARMBSC0 nucleic acid parameters [56]. After a 50 000-step energy minimization procedure, the systems were heated and equilibrated for 1 ns in the NVT ensemble at 310 K and 323 K, followed by a 200 ps pre-NPT equilibration using the Berendsen barostat [57] and a subsequent 1 ns NPT equilibration using the Parrinello-Rahman algorithm [58] at 1 atm. To investigate the dynamic basis of thermostability, MD simulations were performed on apo-protein systems at 310 K using the CHARMM27 force field [59]. For these systems, the NPT equilibration was conducted via a single 1 ns phase directly following the NVT ensemble, omitting the pre-equilibration step used for the complex systems.

All systems were solvated in cubic water boxes containing 0.15 M NaCl to mimic the experimental conditions. Subsequent 200 ns production simulations were executed with periodic boundary conditions and the integration step was set to 2 fs. The LINCS algorithm was employed to constrain covalent bonds involving hydrogen atoms [60]. Lennard-Jones interactions were truncated at 10 Å. The electrostatic interactions were calculated using the particle mesh Ewald method with a cut-off of 10 Å on an $\sim$ 1.6 Å grid with a fourth-order spline [61]. Temperature control was maintained using the velocity rescaling thermostat [62]. All simulations were performed using GROMACS 2025.1 software packages.

Binding free energies for the protein-nucleic acid (DNA and RNA) complexes were quantified using the Molecular Mechanics Poisson-Boltzmann Surface Area (MMPBSA) method [63]. The analysis utilized the final 100 ns of pre-computed MD trajectories, ensuring the system had reached conformational equilibrium. To identify and quantify the contribution

of key residues to intermolecular binding, a residue-wise binding energy decomposition was performed. This decomposition employed the ‘decoupling’ strategy integrated within the gmx-MMPBSA v1.5.x packages.

Results

Dual-fitness evolution strategy to redesign T7 RNA polymerase for thermostability and high-temperature activity

To navigate the complex fitness landscape of a given protein, we established a hybrid prediction framework encompassing both single-site and multi-site mutation strategies. For single-site exploration, we utilized a pre-trained PLM to conduct unsupervised prediction across all possible single-site saturation mutations. Regarding multi-site prediction, we implemented an active learning strategy for supervised prediction, as shown in [Supplementary Fig. S4](#). Specifically, we employed the pre-trained model to extract sequence features for multi-site variants and used an MLP as the regression model to predict the relationship between features and fitness.

In order to achieve simultaneous optimization of dual-fitness, we implemented a co-evolutionary workflow within our modeling framework. This strategy is operationally achieved through parallel supervised training of two specialized fitness prediction models, with its detailed implementation systematically organized into six key steps ([Fig. 1](#)). In the first step, single-site mutants are scored by a pre-trained PLM using its zero-shot prediction functionality (detailed in the Materials and methods), with the top 30–100 mutants selected for wet-lab evaluation. Following dual-fitness evaluation, the entire architecture is fine-tuned using the wet-lab evaluation data (detailed in the Materials and methods). We then utilized the fine-tuned model to score and rank multi-site combinations. Due to the combinatorial explosion of possible mutations, combining all ranked sites is computationally intractable; thus, we focused on recombinations of the top 10–15 sites to balance search space coverage with computational efficiency. We address the challenge of combining dual-fitness effects by using one fitness for ranking and the other as a cut-off, ensuring a balanced selection criterion. In each round, the top 20–50 ranked multi-site mutants are selected for further wet-lab testing. The number of sites combined could be gradually increased in subsequent rounds as more multi-site data become available, enhancing the model’s predictive accuracy. Such an iterative cycle continues until an optimal mutant is identified.

T7 RNAP was selected as the validation target due to the industrial imperative for simultaneous optimization of thermostability and high-temperature activity, particularly for reducing dsRNA by-products [46, 64]. Initially, the pre-trained PLM PRIME was used to score and rank the single-site saturating mutations of T7 RNAP, and the top 55 single mutants were selected for the experimental evaluation. As shown in [Fig. 2a](#), the T_m and 37°C activity of the mutants were tested. Among all single-site mutants tested, 27 of them exhibited a higher T_m compared with the WT. Additionally, 46 mutants retained at least 40% of the WT activity at 37°C. Given the limited enhancement of protein stability by single amino acid changes, and that none of the T_m values exceeded 49°C, we assessed the residual activity of these mutants with higher T_m by thermal incubation at 48°C for 5 min to serve as a

fitness screening metric instead of high-temperature activity ([Supplementary Fig. S5](#)).

Following the experimental validation, we evaluated the model’s discovery potential by cross-referencing these 54 characterized mutations against 208 previously reported sites curated from a comprehensive literature survey ([Supplementary Table S2](#)) [39, 48, 65–74]. Notably, 75.9% (41 sites) of our identified candidates are distinct from these historically established sites, while the remaining 24.1% (13 sites) overlap. Crucially, our selection process relied exclusively on autonomous model predictions without prior knowledge of these literature hotspots. This high rate of independent discovery underscores the potential of deep learning to identify non-intuitive functional sites and expand the known mutational landscape of T7 RNAP.

For the second round, we selected 12 single-site mutations (M306K, T375K, L446F, S430P, A468F, P476E, N601E, S606V, P657K, Q786L, W797L, and Y846R) to construct a combinatorial library. Lacking the data for high-temperature activity, we used the T_m data from 54 mutants to fine-tune PRIME. Multi-site variants derived from these 12 mutations were then scored and ranked, with 40 selected for experimental validation: 20 two-site, 5 three-site, 5 four-site, and 10 eight-site mutants. All multi-site mutants were expressed, purified, and characterized for their T_m and 52°C activities. [Figure 2b](#) illustrates a notable increase in T_m for the second round of multi-site mutants compared with the single-site mutants, with the median T_m increased by ~5°C. Concurrently, as shown in [Fig. 2c](#), the multi-site mutants began to exhibit detectable activity at 52°C. Considering both T_m and 52°C activity, the four-site mutant performed best, while the eight-site mutants suffered from significant activity loss despite a larger increase in T_m ([Supplementary Fig. S6](#)).

The results from the initial two rounds of experiments were leveraged to further fine-tune the model. Specifically, all accumulated data were used to train the T_m predictor, while the data from the second round of multiple sites were used for the 52°C activity predictor. In addition to the increase in fine-tuned data compared with the second round, we added eight new single-site mutants due to their good thermostability or activity performance (I117L, I217L, S397W, C510E, S633P, K642G, A703T, and P742G) to the combination in the third round, adjusting the multi-site range from 2–8 sites to 4–8 sites. A total of 55 multi-site mutants were selected for experiments in the third round, comprising 10 four-site, 9 five-site, 5 six-site, 11 seven-site, and 20 eight-site mutants. As shown in [Fig. 2b](#) and [c](#), multi-site mutants from the third round continued to exhibit a steady increase in both the median and maximum T_m compared with the previous two rounds. However, although the third round displayed higher maximum and average activities compared with the second round, it also had a lower minimum activity at 52°C, resulting in a broader distribution. Although integrating high-temperature activity data into model fine-tuning proved effective, the resulting performance exhibited variability and limited robustness.

With the data from the third round of experiments, we further refined the model for the fourth round. In this round, multi-site mutations were aggressively limited to 6–10 sites, and a total of 30 multi-site mutants were selected, comprising 5 six-site, 6 seven-site, 7 eight-site, 6 nine-site, and 6 ten-site mutants (see [Supplementary Table S3](#) for details). With the addition of more multi-site mutation data for training, the new mutants demonstrated greater thermostability, with

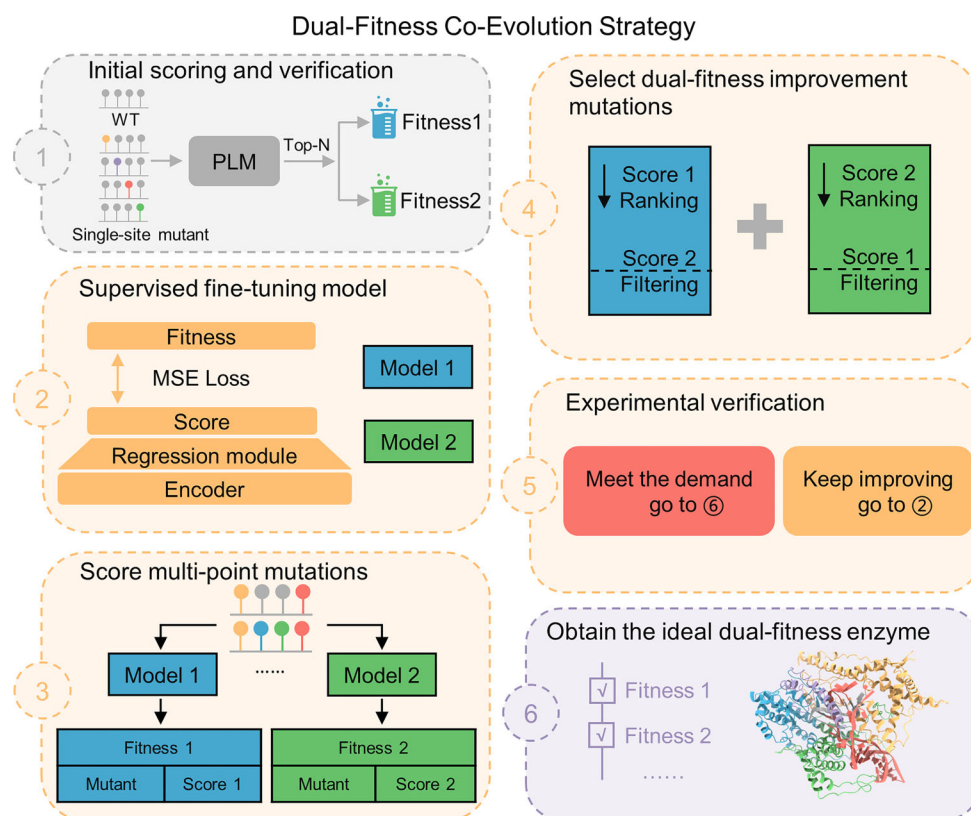


Figure 1. Overview of the dual-fitness co-evolution strategy. In step 1, top-N mutations are generated by a pre-trained PLM in a zero-shot manner, with scoring based on log odds calculations between mutant and WT sequences, as detailed in the Materials and Methods. These selected mutants are experimentally verified to obtain fitness 1 and fitness 2, which are used to fine-tune the model in step 2. Step 3 utilizes the two fine-tuned models to score the combination of mutations. Step 4 performs ranking and filtering to select the mutants that improve dual-fitness simultaneously. Step 5 characterizes the mutants to determine if they meet application demands, and the final step, step 6, characterizes the ideal dual-fitness protein. The steps for single-site mutation prediction/validation, multi-site mutation prediction/validation, and comprehensive experimental characterization are represented by gray, orange, and purple backgrounds, respectively.

an overall improvement in T_m compared with the previous round. Although the distribution of activity at 52°C remained broad, the average activity showed a notable increase (Fig. 2b, c).

Data from the first four rounds of experiments provided sufficient training samples to refine the model, enabling it to explore a broader fitness landscape. In the fifth round, we generated 9–14 site mutants by combining all prior single-site mutations, scoring and ranking them with the updated model. The top 40 ranked mutants were selected for experimental validation, comprising 14 nine-site, 11 ten-site, 5 eleven-site, 2 twelve-site, 3 thirteen-site, and 5 fourteen-site mutants. Experimental validation demonstrated that the fifth-round multi-site mutants exhibited not only high stability and enhanced high-temperature activity but also a more concentrated fitness distribution. As shown in Fig. 2b and c, the median T_m reached 55°C while the median activity at 52°C rose to 50%, confirming successful co-evolution of dual-fitness attributes.

Overall, each round of evolution substantially improved both the T_m and 52°C activity of T7 RNAP. After five rapid iterations, the median T_m increased by ~10°C compared with the single-site mutants, and by ~5°C compared with the multi-site mutants in the second round. The median high-temperature activity at 52°C also rose from ~5-fold to 40-fold compared with that of the WT, which exhibits only ~1% of its activity at 37°C when measured at 52°C. Remarkably, the best-performing mutant exhibited >60 times the activity

of the WT at 52°C and attained ~65% of the performance of the WT at 37°C. Details of all mutations are presented in [Supplementary Tables S3–S5](#).

In addition to the median increase, both T_m and 52°C activities show a more concentrated overall distribution and more stable performance, indicating high predictive reliability. With >100 data points accumulated, the fine-tuned model demonstrated notably improved accuracy for multi-site predictions ([Supplementary Figs S7 and S8](#)). While this study utilized five rounds to systematically demonstrate the trajectory of deep learning-guided evolution, the rapid convergence observed in later rounds suggests that this strategy can efficiently accelerate protein redesign campaigns with minimal experimental burden.

Validation of multi-fitness T7 RNA polymerase mutants for applications

From the fifth round, we selected 11 multi-site mutants with $T_m > 54^\circ\text{C}$ and 52°C activity >45% (relative to the WT at 37°C) for further validation under more demanding application conditions. These mutants were subjected to reactions at 52 and 55°C for 3 h, respectively. The top performers advanced to further analytical assays, with results presented in Fig. 3a. Notably, a 14-site mutant M1 (A124N, I217L, T375K, M369T, S397W, S430P, L446F, N529V, L534V, S606V, S633P, G618E, L665D, and Q786L) demonstrated exceptional ther-

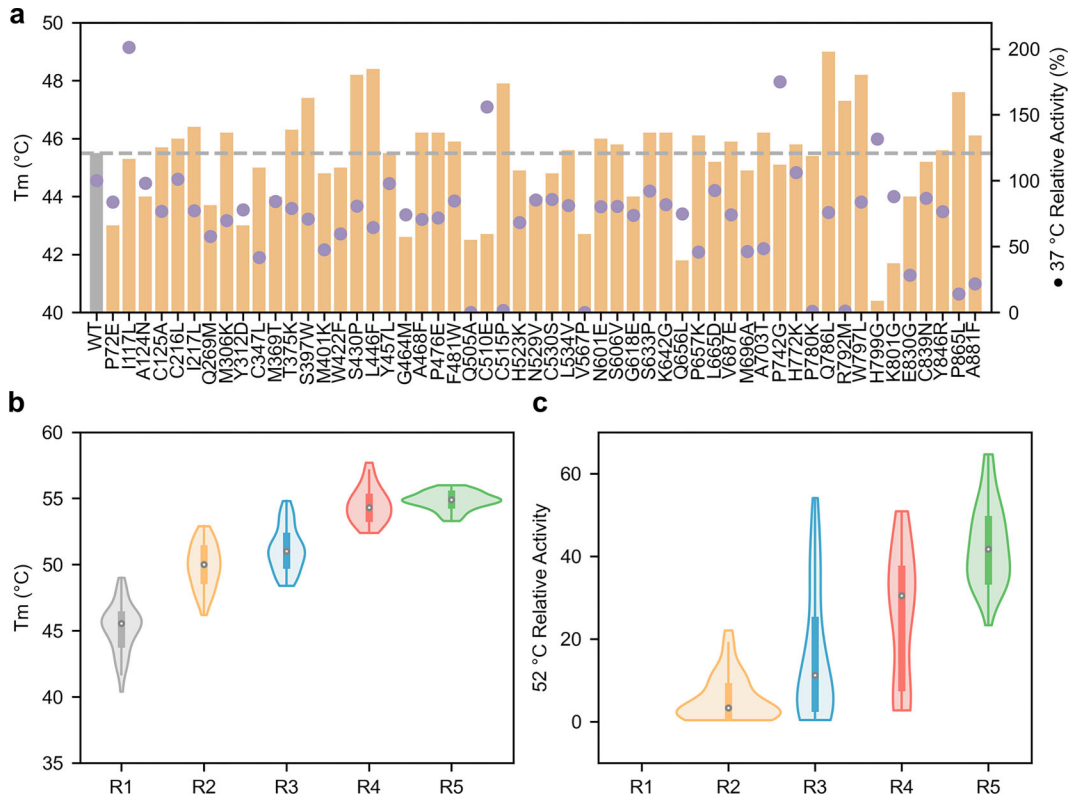


Figure 2. The experimental results of mutants. (a) The T_m (bar) and 37°C relative activity (scatter) for all single-site mutations. WT is colored in gray, and the mutants are colored in orange. The dashed line indicates the T_m of the WT. Violin plots show the distribution of T_m (b) and 52°C relative activity (c) for each evolution round. Relative activities are calculated based on the 37°C activity of the WT. Rounds 1–5 are represented by R1–R5 and colored in gray, orange, blue, red, and green, respectively.

mostability and activity, outperforming other candidates and highlighting its potential for high-temperature applications.

Using far-UV CD variable-temperature sweeps, the changes in secondary structures of the WT and M1 between 25 and 65°C were detected. As shown in Fig. 3b, the WT has an onset unfolding temperature of ~40°C and a T_m of 46.9°C. In contrast, M1 has an onset unfolding temperature of >50°C. Crucially, at temperatures where the WT was fully unfolded, M1 retained a well-ordered secondary structure, exhibiting a T_m increase of ~10°C compared with the WT. Additional DSC measurements also confirmed that M1 has a T_m 11°C higher than that of the WT, consistent with the previous results (Fig. 3c). Combining the results of CD, DSC, and differential scanning fluorimetry (DSF), we can conclude that the stability of M1 is substantially improved compared with the WT.

Subsequently, we specifically characterized the activity of M1 by conducting the reaction at 37, 40, 42.5, 45, 47.5, 50, 52.5, 55, and 57.5°C. As shown in Fig. 3d, M1 shows catalytic activity in the temperature range of 37–57.5°C, with the optimal catalytic temperature range of 45–50°C, where it retained ~80% of WT activity at 37°C. However, at 50°C, the WT retains merely 5% of its activity observed at 37°C (Supplementary Fig. S9). In addition to using fluorescence assay to measure the activities of the WT and M1, we also utilized longer DNA sequences as transcription templates and purified the mRNAs to determine the mRNA yields at different temperatures. As depicted in Fig. 3e, the transcription efficacy of M1 was consistent across both long (~1000 bp) and short (~100 bp) templates. Notably, the transcription efficacy of M1 for the long template remains high between 37

and 50°C, maintaining 80% of the transcriptional activity of the WT observed at 37°C. These findings demonstrate that our dual-fitness evolution strategy yielded a T7 RNAP mutant with exceptional thermostability and high-temperature activity, ideal for applications requiring transcription at elevated temperatures.

Focusing on the application scenario of mRNA production, we quantified dsRNA by-product content using ELISA as a critical third fitness metric on M1. Interestingly, we found that while increasing the temperature from room temperature to 42.5°C decreased the proportion of dsRNA for a 1000 bp template, further temperature increases did not reduce dsRNA content (Fig. 3f). Additionally, the proportion of by-products from the reaction at 52.5°C was similar to that obtained from the WT reaction at 37°C, indicating that higher temperatures do not necessarily reduce by-product formation. Balancing yield and product purity, we chose 50°C as the optimal reaction temperature for subsequent experiments. We prioritized the top six mutants from 11 better-performing mutants based on their thermostability and activity at 52 and 55°C, evaluating their transcription yield and by-products based on the 1000 bp template. Among the screened candidates, mutants M8, M10, and M11 matched M1's yield but produced only ~30% of M1's dsRNA at 50°C, a substantial improvement over the WT (Fig. 3g). Furthermore, we examined the kinetic stability of these three mutations in T_{50}^{15} . As shown in Fig. 3h, we found that M10 and M11 had improved kinetic stability, with an increase of ~10°C compared with the WT. We also explored the transcriptional effects of M10 and M11 on longer templates and found that these mutants showed en-

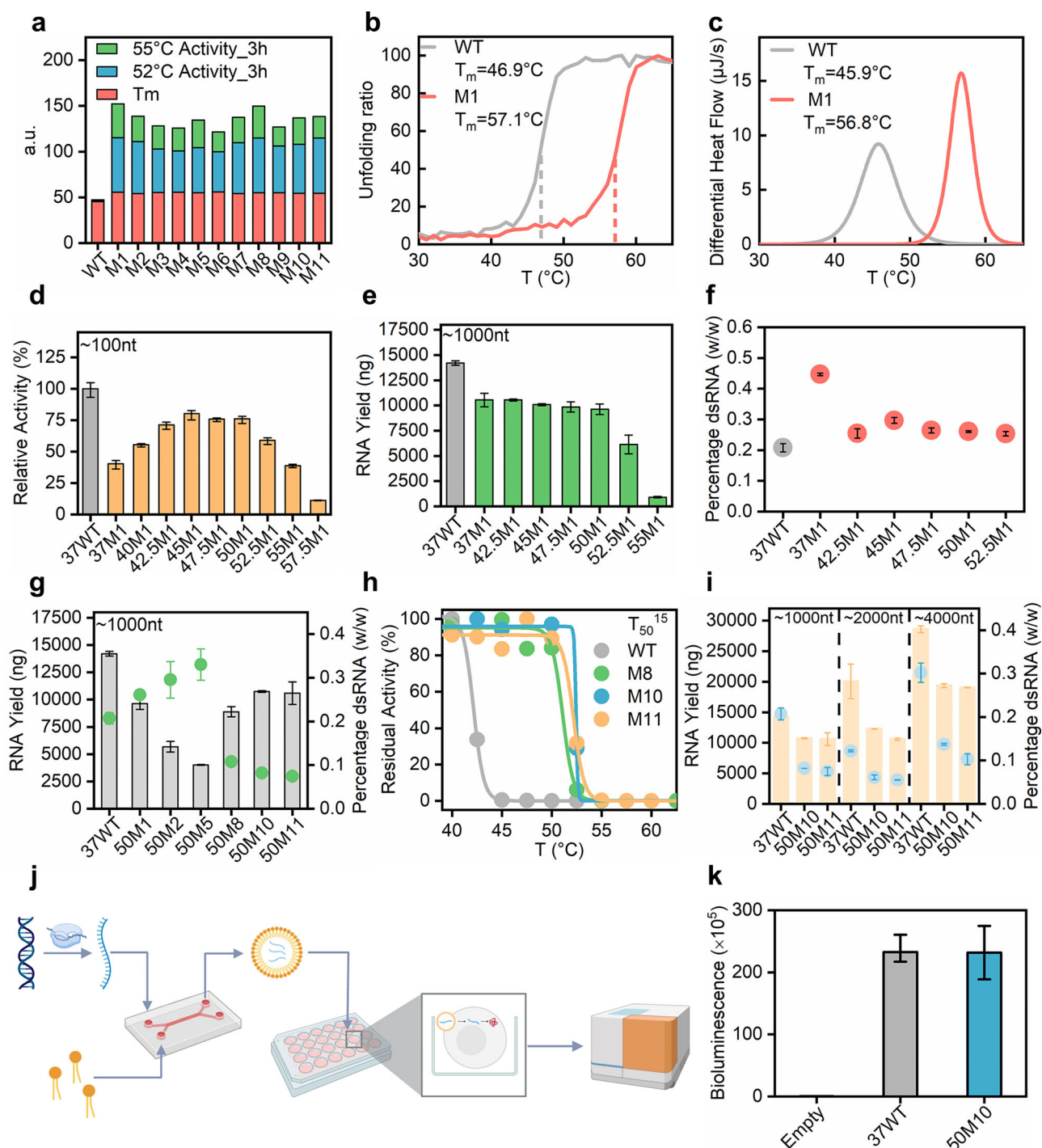


Figure 3. Comparison between WT and mutant T7 RNAP. **(a)** Selection of the improved dual-fitness mutant. Comparison of thermal unfolding curves of the WT and M1 determined by **(b)** far-UV CD spectroscopy and **(c)** DSC. Effects of temperature on activity of M1 under ~100 bp **(d)** and ~1000 bp **(e)** templates. **(f)** The dependence of dsRNA impurities for M1 on temperature. **(g)** The RNA yield (bar) and dsRNA impurities (scatter) under the ~1000 bp template for the WT and mutant. **(h)** The kinetic stability of the proteins is determined by monitoring residual enzymatic activity after incubation for 15 min at various temperatures. The T_{50}^{15} values for the WT, M8, M10, and M11 are 42.1, 51.1, 52.4, and 52.2°C, respectively. **(i)** RNA yield (bar) and dsRNA content (scatter) of the WT, M10, and M11 under different length templates. **(j)** Diagram illustrating the process from transcription to preparation of mRNA-LNPs, followed by cell transfection and characterization of expression efficiency. **(k)** The expression efficiency in HEK293T cells of mRNA encoding luciferase transcribed by the WT at 37°C and M10 at 50°C. The relative activity shown in **(a)** and **(d)** is based on the WT at 37°C. The residual activity shown in **(h)** is based on the maximum activity of each protein.

hanced activity for the length of templates of ~100, 1000, 2000, and 4000 bp. Regardless of template lengths, both mutants demonstrated a great reduction in by-products compared with the WT (Fig. 3i).

To further validate the industrial applicability of the mutant, we conducted cell transfection experiments as depicted in the schematic diagram (Fig. 3j). We transcribed the luciferase template (~2000 bp) with the WT and M10, and encapsulated the transcribed mRNA in a microfluidic device to form mRNA-LNP complexes. The encapsulation efficiency was >97%, with the particle size ~120 nm, and the PDI was 0.2 (Supplementary Table S6). The luciferase mRNA-LNPs were transfected into HEK293T cells and incubated for 24 h. As shown in Fig. 3k, luciferase expression levels from mRNA synthesized by M10 at 50°C were comparable with those from WT-synthesized mRNA at 37°C, confirming the functional integrity of the product. Overall, comprehensive characterization identified M10 as the optimal mutant, demonstrating excellent thermostability (Supplementary Fig. S10), high-temperature activity, low by-product content, and efficient mRNA product transfection. This indicates that our strategy has successfully provided a multi-fitness evolved mutant based on the WT.

Evolutionary trajectories and the emergence of epistasis in protein mutants

The multi-fitness enhancements observed in our evolved mutants prompted a systematic analysis of their evolutionary trajectories. The intrinsic correlations between activity and stability, alongside the epistatic complexity inherent in the evolutionary process, contribute to a rugged fitness landscape that underscores the model's sophisticated navigational logic.

Analysis of variants from R2 to R5 revealed a complex interplay between thermostability and catalytic performance at different temperatures. We observed an inherent trade-off between T_m and ambient activity ($\rho = -0.37$, Fig. 4a; Supplementary Fig. S11), reflecting a predictable upward shift in the optimal catalytic temperature (T_{opt}) as evolutionary pressure focused on high-temperature performance rather than ambient efficiency (Supplementary Fig. S9). Interestingly, while the global thermal profile shifted, ambient and high-temperature activities remained positively correlated overall ($\rho = 0.32$, Fig. 4b). Specifically, this correlation strengthened substantially as thermostability increased: the Spearman's ρ rose from 0.60 for mutants with a T_m of 49–52°C, to 0.74 for those between 52 and 55°C, and reached 0.85 for variants exceeding 55°C (Supplementary Table S7). This trend suggests that catalytic competence acts as a global effect: once structural stability is no longer a limiting factor, variants with higher intrinsic activity at ambient temperatures inherently exhibit superior performance at elevated temperatures. Ultimately, the positive correlation established between T_m and activity at 52°C ($\rho = 0.58$, Fig. 4c) confirms that stability serves as the indispensable foundation for high-temperature function; more importantly, it validates our co-evolution strategy, which successfully synchronized structural reinforcement with functional optimization.

A critical question arises: why does catalytic activity manifest as a global effect, yet exhibit divergent correlations with thermostability across different temperatures? These relationships are governed by the upward shift of the T_{opt} , which currently lies bracketed between 37 and 52°C

(Supplementary Fig. S9). As T_m increases, the rightward shift of the T_{opt} curve moves the 52°C point closer to the peak while pushing the 37°C point further down the ascending limb, creating an inherent trade-off. However, because catalytic competence is also a global attribute, any change in the curve's peak height (intrinsic V_{max}) scales activity across all temperatures simultaneously. This explains why thermostability, while providing the essential physical foundation for high-temperature function, does not guarantee it. Ultimately, the interplay between T_{opt} shifts and intrinsic efficiency creates a rugged co-evolutionary landscape, necessitating the precise synchronization of stability and activity.

To further characterize the evolutionary complexity, we performed global and local analyses of epistasis—the non-additive interactions where the combined effect of mutations deviates from the linear sum of individual contributions. Traditional directed evolution, which often relies on the stepwise accumulation of beneficial mutations, frequently fails to account for these interactions. We evaluated this by comparing the measured thermostability of multi-site mutants against the linear additive results derived from individual mutations (Fig. 4d). The substantial deviation of most data points from the diagonal identity line ($y = x$) indicates the prevalence of epistatic effects. Notably, mutants from initial rounds were dominated by antagonistic epistasis (clustering below the diagonal), whereas variants exhibiting potent synergistic epistasis (shifting toward or above the diagonal) emerged exclusively in the final round. This progressive transition from antagonism to synergy demonstrates the capacity of our strategy to harness positive multi-residue interactions that elude simple stepwise methods (Supplementary Figs S7 and S12).

We further dissected these interactions through a local analysis of mutation frequency and specific combinatorial pathways (Fig. 4e, f). Frequency analysis revealed that superior individual thermostability did not guarantee inclusion in optimal combinations, while some frequent mutations might display only modest individual stability. For instance, W797L exhibits excellent single-site stability and maintains prevalence in intermediate rounds; however, it fails to synergize within higher-order mutational contexts, leading to its absence from the final optimal variants. Conversely, G618E—despite its poor single-site stability—is retained in the final optimized mutants due to its strong positive epistatic contributions when combined with other key mutations. This discrepancy highlights that protein evolution is a complex, multi-dimensional optimization process driven by dual selection pressures for thermostability and activity, and further modulated by epistatic interactions. While these mutations were ultimately selected based on overall fitness, their prevalence in the population identifies them as key evolutionary nodes. To ensure a clear visualization of the evolutionary trajectory, we focused on the thermostability dimension.

Network analysis explained this phenomenon by revealing complex local landscapes: simple combinations of individually highly stable mutants (e.g. S430P combined with L446F) frequently resulted in antagonistic effects, yielding lower stability than their components (Fig. 4f). Despite this local antagonism, the retention of both S430P and L446F in the final variants suggests that they provide a crucial structural context that accommodates further beneficial mutations, highlighting the non-additive and rugged nature of the fitness landscape. In contrast, 'weak' single-site mutants often acted as crucial potentiators; for example, T375K ($\Delta T_m = 0.8^\circ\text{C}$), despite be-

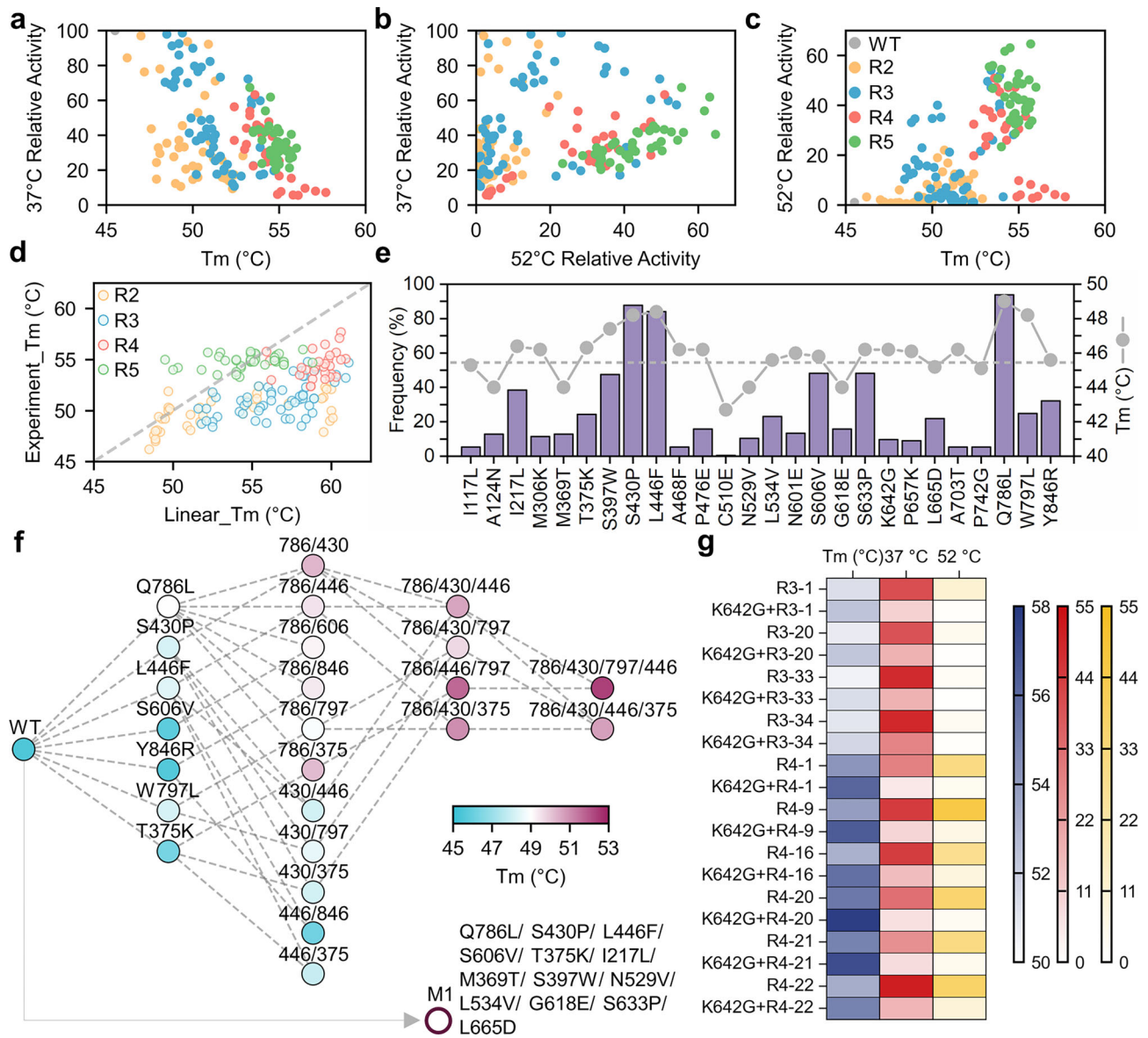


Figure 4. Evolutionary and epistasis analysis of mutational effects on protein fitness. (a–c) Evolutionary trends: (a) correlation between T_m and catalytic activity at 37°C; (b) comparison of catalytic activity for mutants at 37 and 52°C across four rounds of evolution; and (c) correlation between T_m and activity at 52°C. (d–f) Epistatic effects on thermostability. (d) Evaluation of epistasis by comparing the measured thermostability of multi-site mutants with a linear prediction from individual mutations. The dashed line serves as the reference line $y = x$. (e) The frequency and thermostability of single-site mutations combined in multi-site mutants. The dashed line corresponds to the T_m of the WT. (f) The evolutionary pathway of the most frequent single-site mutant. Nodes represent mutants, and edges represent mutation pathways. Note that color depth corresponds to the T_m of each mutant, illustrating the role of epistasis. (g) The epistatic effects on both thermostability and activity, focusing on the K642G single-site mutation and its influence on both fitness traits.

ing ineffective with S430P, synergized with Q786L to identify high-stability combinations (visualized as darker nodes in Fig. 4f) that might be missed by greedy selection strategies. These findings demonstrate that optimizing thermostability requires navigating a complex epistatic landscape—a task efficiently achieved by our learning-based evolutionary strategy rather than simple linear combination.

Systematic analysis of the fitness landscapes (Supplementary Figs S13–S15) identified K642G as a representative mutation that exemplifies the complexity of multi-fitness epistasis. The individual impact of K642G appeared negligible, as it produced only a marginal increase in thermostability ($\sim 0.7^\circ\text{C}$) alongside a minor decrease in

activity ($\sim 80\%$ of the WT). However, multi-site mutations containing K642G universally exhibited high thermostability but depressed activity, regardless of the assay temperature (37 or 52°C). To further characterize this behavior, we compared the fitness changes of mutants with and without K642G, as shown in Fig. 4g. The inclusion of K642G in multi-site backgrounds consistently triggered a ‘stabilizing yet silencing’ effect: it boosted thermostability by 1–3°C but universally depressed activity at both 37 and 52°C. This finding reveals a striking layer of complexity: a mutation with minimal individual impact can exert profound, trait-specific epistatic effects when entangled with other residues. This highlights that in multi-fitness evolution, the epistatic landscape is not merely

rugged but coupled, necessitating advanced learning-based strategies to manage these intricate trade-offs.

Structural and dynamic basis for improved protein fitness

To move beyond evolutionary statistics and uncover the structural origins of enhanced fitness, we sought to elucidate the molecular basis of increased thermostability and activity (Fig. 5). By transitioning from global thermodynamic assessments of free energy and binding affinities to a local dissection of residue-level interactions, we systematically evaluated both single-site and multi-site mutants to isolate the distinct physical mechanisms underlying these traits.

We utilized FoldX [15, 75] to estimate mutation-induced free energy changes; however, predicted $\Delta\Delta G$ values showed poor correlation with experimentally measured thermostability (Supplementary Fig. S16a). Notably, several mutations displayed clear contradictions between prediction and experiment: the most thermostable variant, Q786L, was predicted to be destabilizing, while both experimentally destabilizing (L665D and G618E) and stabilizing (L534V and K642G) mutations were uniformly classified as unfavorable by FoldX. Moreover, mutations predicted to be destabilizing were readily incorporated into multi-site variants that ultimately exhibited high overall thermostability. To assess whether local interaction networks could explain these discrepancies, we systematically analyzed hydrogen bonds, salt bridges, and hydrophobic contacts for individual mutants (Supplementary Fig. S16). While some substitutions followed conventional stabilization principles [76–80], others exhibited stability gains or losses that defied explanation by local interaction changes, solvent accessibility, or secondary structure context (Fig. 5a; Supplementary Fig. S17). These observations indicate that local energetic and structural features alone do not directly determine thermostability in this system, providing a plausible explanation for the systematic failure of FoldX and related heuristics. Given that a comprehensive mechanistic description remains elusive, our sequence-based strategy offers a more robust and effective alternative for optimizing thermostability, without relying on explicit assumptions about local interaction networks.

Extending our investigation from static structures to dynamic behavior, we performed MD simulations to compare the conformational profiles of the WT and the top-performing multi-site mutant M1. As shown in Fig. 5b, mutant M1 exhibited consistently lower root mean square deviation (RMSD) values than the WT throughout the simulation. This sustained reduction in RMSD indicates enhanced global structural rigidity, suggesting that the mutant's stability derives from a more constrained conformational ensemble. Further supporting this stiffening effect, root mean square fluctuation (RMSF) analysis (Fig. 5c) reveals that residues in M1 generally exhibit smaller fluctuations across the sequence compared with the WT. Importantly, the RMSF values show particularly minimized fluctuations in the regions surrounding residues 220, 390, and 600. These regions spatially correspond to the key mutation sites (217, 375, 397, and 606), suggesting that these targeted mutations rigidify local subdomains, propagating stability across the global fold.

Investigating catalytic activity is inherently complex due to the multi-domain architecture of T7 RNAP, where the N-terminal (NTD), thumb, finger, and palm domains coordinate

the catalytic cycle from initiation to elongation (Fig. 5d, e) [54, 55, 81, 82]. The transition between these phases involves extensive conformational rearrangements, particularly within the NTD, which manages promoter recognition and exit channel formation.

To navigate this functional complexity, we first evaluated the 54 validated mutations through the lens of evolutionary conservation. While mutations at highly conserved sites are typically avoided in protein engineering to minimize functional disruption, our systematic mapping—using 177 homologous sequences—revealed that the vast majority of our candidates (85.2%, 46/54) are located at non-conserved sites (Supplementary Table S8). In contrast, only 14.8% (8 sites) reside in highly conserved positions (Q269M, W422F, A468F, C515P, M696A, P780K, E830G, and P865L). Notably, among the 12 mutations that substantially modulated activity, four within conserved regions (C515P, P780K, E830G, and P865L) led to marked functional impairment. Conversely, the remaining high-impact mutations were distributed across non-conserved regions, suggesting that our model successfully identifies evolvable hubs within these plastic domains.

To interpret the mechanistic impact of these specific mutations, we mapped their spatial dynamics relative to the active site (D537, K631, and D812) (Fig. 5f). Structural superposition revealed that the transition from initiation to elongation is driven by a massive displacement of the NTD. Notably, I117L, located in the NTD, exhibits the most substantial spatial displacement between the two phases. This distal mutation may facilitate NTD rearrangement by alleviating local packing constraints during the initiation-to-elongation transition. Conversely, mutations such as A881F, situated proximal ($\sim 4\text{--}5$ Å) to the active site, probably impose direct steric hindrance, reducing pocket volume and impairing catalysis. This observation is consistent with established research emphasizing the critical role of the T7 RNAP C-terminal tail in stabilizing both NTPs and the nascent RNA transcript [54]. As other sites are positioned > 8 Å from the active center, their influence probably stems from indirect interaction networks.

We further quantified the contributions of key residues to DNA/RNA binding using residue contact maps and MMPBSA decomposition (Fig. 5g; Supplementary Figs S18–S20).

- **Direct contact effects:** P780K and P742G are situated within the finger domain, a region critical for stabilizing the template–product heteroduplex. Specifically, the P780K mutation replaces a rigid proline with a positively charged, long-chained lysine. This substitution probably disrupts the α -helix kink and introduces non-native electrostatic interactions or steric clashes that destabilize substrate binding, leading to the observed functional impairment. In contrast, P742G replaces a constrained proline with a flexible glycine, alleviating steric strain and facilitating smoother conformational transitions during substrate binding, thereby boosting activity.
- **Indirect network effects:** distal mutations such as R792M and C510E, while lacking direct DNA/RNA contact, engage in extensive interaction networks with over five key residues. Perturbations at these hubs are likely to trigger allosteric ‘ripple effects’ that propagate to the active site, reducing overall affinity.
- **Rigidity–activity trade-off:** the mutation C515P enhances rigidity (raising the T_m by 2°C) but abolishes activity, exemplifying the classic stability–function trade-

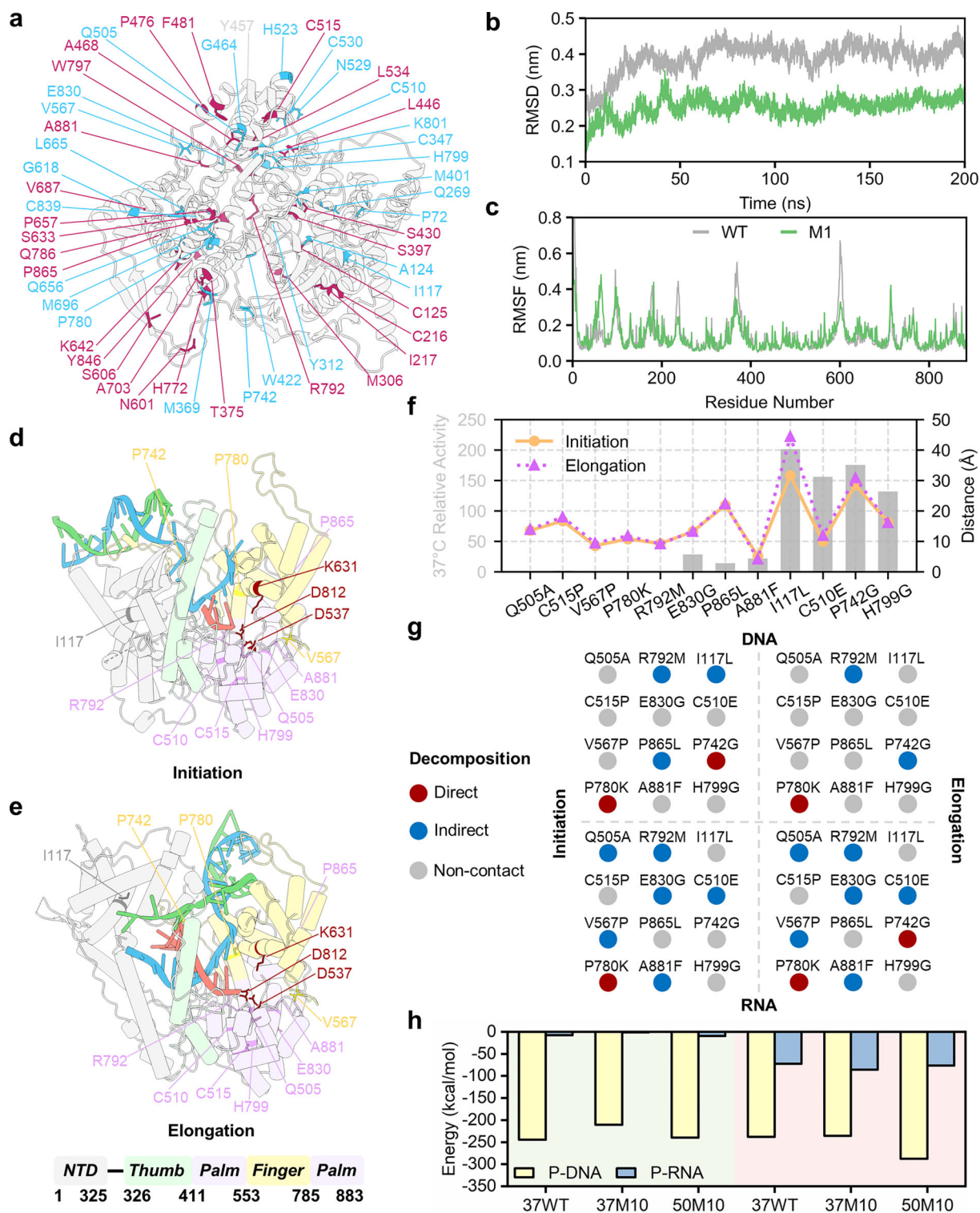


Figure 5. Mechanistic insights into the structural and dynamic determinants of protein fitness. (**a**) Spatial distribution of single-site mutations mapped onto the WT apo structure (gray cartoon). Stability-increasing sites are shown as red sticks, whereas stability-decreasing sites are shown as blue sticks, and sites with no change in stability are shown in gray. MD simulation results comparing the conformational dynamics of WT and M1 (gray and green, respectively), using RMSD (**b**) and RMSF (**c**) to evaluate overall structural stability and residue-level flexibility. Locations of 12 single-site mutations with notable effects on activity across the initiation (**d**) and elongation (**e**) phases of transcription. The T7 RNAP domains (NTD, thumb, palm, and finger) are distinguished by color (gray, green, purple, and yellow), with key catalytic residues (D537, K631, and D812) highlighted in red sticks. The template DNA, non-template DNA, and RNA are shown in blue, green, and pink, respectively. (**f**) The activity of these influential mutants (gray bars) alongside the distance change from their respective mutation sites to the catalytic center during the two transcription stages (represented by orange and purple lines). (**g**) Residue interaction map of the WT with DNA and RNA in initiation and elongation stages. Residue-wise binding energy decomposition was performed to identify critical sites that interact with DNA and RNA. The map visualizes the impact of key mutations on activity, with interaction status indicated by colored markers: red (direct contact), blue (indirect contact), and gray (no contact). (**h**) Comparative substrate-binding free energies of multi-site mutant M10 relative to the WT during the initiation (green background) and elongation phases (pink background).

off. Conversely, H799G disrupts local salt bridges, sacrificing stability for enhanced flexibility that promotes transcriptional turnover.

- Global protein dynamics: the P865L mutation increases loop disorder, potentially impairing catalytic precision. Terminal mutations V567P and E830G are thought to perturb interactions with C-terminal residues 881/882, reducing nascent RNA synthesis.

To quantify the thermodynamic basis of enhanced activity in the multi-site mutant M10, we calculated substrate binding free energies relative to the WT (Fig. 5h). At ambient temperature, the WT exhibits comparable DNA binding during both initiation and elongation, with RNA affinity increasing during the elongation phase. In contrast, M10 shows slightly weaker DNA/RNA binding during initiation but maintains comparable elongation performance, suggesting that its minor reduction in overall activity is primarily initiation limited. However, at 50°C, the scenario inverts. M10 achieves enhanced binding affinity for both substrates across initiation and elongation phases compared with its ambient baseline. Crucially, its binding energetics at 50°C are comparable with those of the WT at 37°C, enabling robust catalytic performance (~80% of the WT).

Beyond stability and catalytic activity, the dynamic regulation of transcriptional fidelity represents a critical dimension of T7 RNAP function. Fidelity is primarily orchestrated by the induced-fit conformational changes during the nucleotide addition cycle (NAC), where the finger domain transitions between open and closed states [83, 84]. This transition serves as a vital checkpoint, utilizing highly conserved O-helix residues—specifically Tyr639 (the 2'-OH discriminator), Phe644 (the geometric filter), and Lys631—to precisely align the substrate for catalysis.

To elucidate how the identified mutations impact fidelity, we performed a systematic mechanistic mapping (Supplementary Table S9). Our analysis reveals that mutations modulate fidelity through three distinct mechanisms: proximal mutations such as P780K (<4 Å from fidelity sensors) directly disrupt the stringent NTP alignment required for the closed state, whereas hinge mutations such as V567P (~13 Å) regulate the flexibility of the conformational pivot during the open-to-closed transition. Furthermore, distal hubs such as C510E and R792M propagate allosteric effects through the protein interaction network, shifting the energetic barriers of the NAC and tuning the stringency of the fidelity checkpoint even in the absence of direct nucleic acid contact.

To investigate the impact of the M10 variant on transcriptional fidelity, we performed MD simulations which revealed a notably more rigid finger domain (reduced RMSE, Supplementary Fig. S21) compared with the WT. This structural rigidification increases the energetic barrier for the 'opening-closing' transition, preventing the 'relaxed' or 'floppy' states that are often prone to misincorporation. Such stabilization is critical for navigating the inherent trade-off between catalytic rate and fidelity [85–87]. While accelerated kinetics typically compromise accuracy by bypassing key checkpoints, M10 operates at 50°C with a catalytic pace comparable with the WT at 37°C. By maintaining this controlled pace within a stabilized framework, M10 effectively avoids the fidelity degradation often seen in hyperactive variants, as further evidenced by its reduced production of transcriptional by-products and preserved functional stringency.

Collectively, these structural, dynamic, and thermodynamic analyses illustrate that the evolved mutants achieve superior performance not through a single mechanism, but rather through a synergistic combination of global stabilization (or targeted flexibility maintenance) and optimized substrate interactions.

Discussion

In this study, we demonstrated the efficacy of a deep learning-guided evolutionary strategy that integrates a pre-trained PLM with iterative experimental validation to simultaneously optimize distinct protein fitness. In contrast to traditional rational design, which relies heavily on detailed structural knowledge and often struggles with complex, distal effects, our approach efficiently identified high-performance variants through data-driven sequence exploration. These variants achieved substantial gains; notably, the top-performing mutant, M1, exhibited a 60-fold increase in high-temperature activity and an enhancement in thermostability (T_m) of >10°C compared to the WT. These results highlight the potential of deep learning-guided strategies to facilitate protein engineering where structural priors are limited or where fitness landscapes are rugged.

A key feature of this strategy is its capacity to account for complex mutational interactions. In our evolutionary trajectory, we observed that the model's predictive robustness and its ability to capture epistatic effects improved substantially as the volume of experimental data and the number of iterative rounds increased. By integrating feedback from successive generations, the PLM was able to prioritize combinations of mutations that might appear neutral or even detrimental in isolation but yielded superior synergistic outcomes. While the internal decision-making of the neural network remains a 'black box', our results suggest that the iterative fine-tuning process allows the model to progressively refine its mapping of the non-linear fitness landscape. This capability facilitates the discovery of complex mutational patterns that are often difficult to resolve through traditional stepwise strategies, which are typically limited in their exploration of multi-site sequence space. However, we note that for the specific challenge of dual-fitness co-evolution, established benchmarks are currently lacking. Consequently, future work should focus on direct head-to-head comparisons with other emerging computational strategies to rigorously evaluate relative performance.

The simultaneous enhancement of thermostability and high-temperature activity provides compelling evidence of the model's efficacy in guiding dual-fitness optimization. As outlined in our results, protein traits often exhibit complex dependencies, such as the relationship observed here where thermostability serves as a structural prerequisite for activity at elevated temperatures. However, fulfilling this prerequisite alone does not guarantee high catalytic efficiency, as rigidification can often compromise turnover. The fact that our strategy consistently identified mutational pathways leading to a coordinated improvement in both traits—rather than yielding variants that were stable but catalytically inert—demonstrates the model's sophisticated guidance in navigating the fitness landscape. By effectively managing the intricate interplay between distinct protein properties and epistatic effects, this data-driven approach facilitates the discovery of synergistic evolutionary routes. This demonstrated the ability to coordinate the improvement of interdependent traits, regardless of

the nature of their interactions, which is crucial for engineering enzymes tailored to satisfy diverse and challenging industrial specifications.

Beyond these primary metrics, the success of polymerase engineering is fundamentally dictated by the selection pressures applied during evolution. While our study prioritized stability and activity, these parameters are intrinsically linked to transcriptional fidelity. Specifically, the minimization of dsRNA by-products serves as a robust functional proxy for enzymatic stringency. Mechanistically, the observed rigidification of the finger domain probably reinforces the induced-fit checkpoint, ensuring the conformational precision required for accurate substrate selection. By co-optimizing these inter-related traits, our framework preserves enzymatic integrity and establishes a strategic foundation for incorporating explicit, high-throughput fidelity assays into future generative pipelines.

Finally, the practical implications of this work are highlighted by the generation of robust T7 RNAP mutants suitable for mRNA vaccine production. Extending beyond the initial dual-fitness goals, our pipeline proved sufficiently flexible to incorporate a third, critical constraint—dsRNA content. The identification of mutants M10 and M11, which maintain high thermostability and activity while reducing immunostimulatory dsRNA by-product levels by ~70%, underscores the versatility of our strategy. These variants not only address the specific limitations of WT T7 RNAP but also serve as promising candidates for industrial-scale high-temperature transcription processes. Ultimately, this work establishes a generalizable framework for engineering enzymes to meet the rigorous and multifaceted demands of modern biotechnology.

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Supplementary data

Supplementary data are available at NAR online.

Conflict of interest

F.J., L.K., Y.Y., and B.W. are named inventors on patent applications (application number: 202410708027.X, granted number: CN117070493B) that have been filed, covering the sequences and applications of the mutants discussed in this manuscript.

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Data availability

All data described are contained within the article.

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